GENERATIVE MODELS

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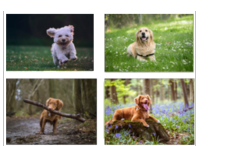
Part I

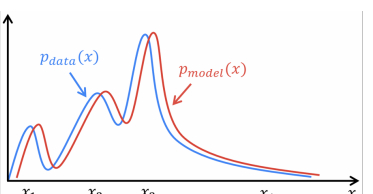
Overview

What is a Generative Model?

1. Assume that the training data $\{\mathbf{x}_i\}_{i=1}^n$ is generated from the probability distribution $p_{\text{data}}(\mathbf{x}) \Rightarrow \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$
2. Train a parameterized model $p_\theta(\mathbf{x}) \Rightarrow p_\theta(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x})$
3. Generate new data from the learned $p_\theta(\mathbf{x}) \Rightarrow \mathbf{x} \sim p_\theta(\mathbf{x})$

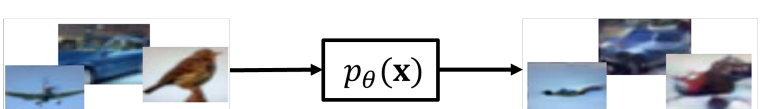


$$\mathbf{x}_i \sim p_{\text{data}}$$
$$i = 1,2,\cdots,n$$



Step 1: density estimation

Step 2: sampling



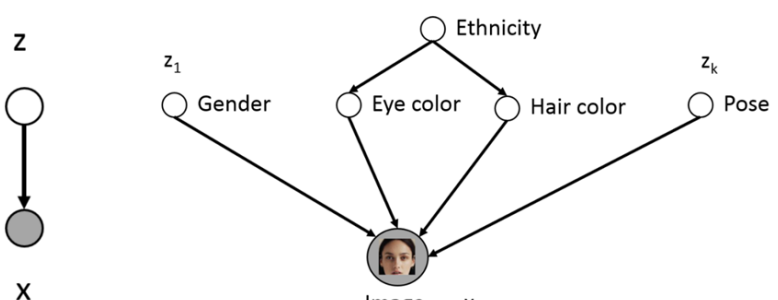
Latent Variable Model

Assume the data has some underlying structure captured by a latent variable $\mathbf{z}$ drawn from a simple distribution. A decoder network then maps $\mathbf{z}$ to $\mathbf{x}$:

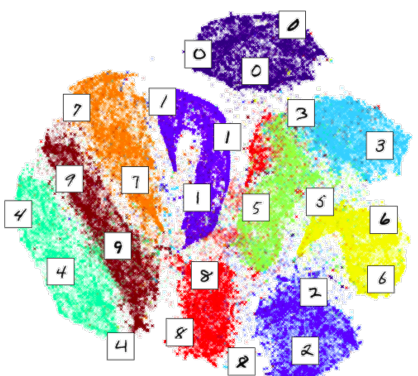
$$\mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I}), \qquad \mathbf{x}|\mathbf{z} \sim p_\theta(\mathbf{x}|\mathbf{z}) = \mathcal{N}(D_\theta(\mathbf{z}), \sigma^2\mathbf{I})$$

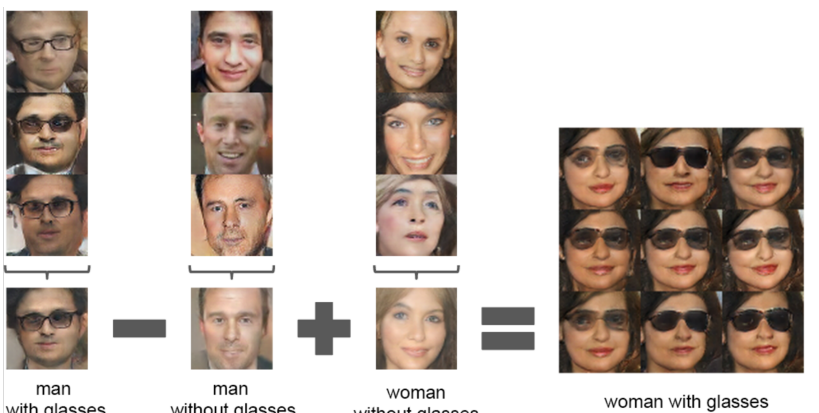
$$p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x}|\mathbf{z})\, p(\mathbf{z})\, d\mathbf{z}$$

Even though $p(\mathbf{z})$ and $p_\theta(\mathbf{x}|\mathbf{z})$ are each simple Gaussians, the mixture over all possible $\mathbf{z}$ can represent complex, multimodal distributions.

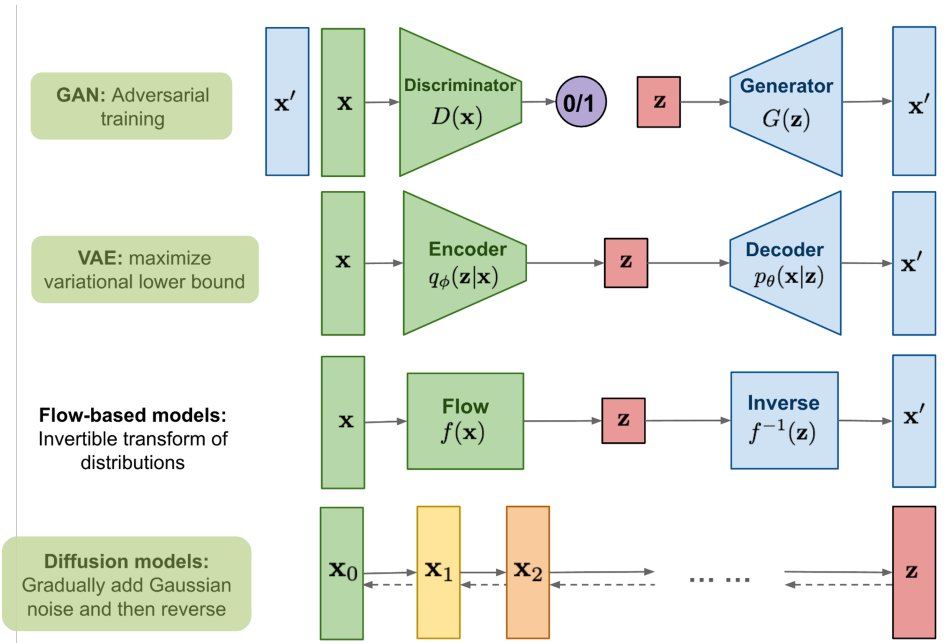


Latent Variable Model





Latent Variable Model

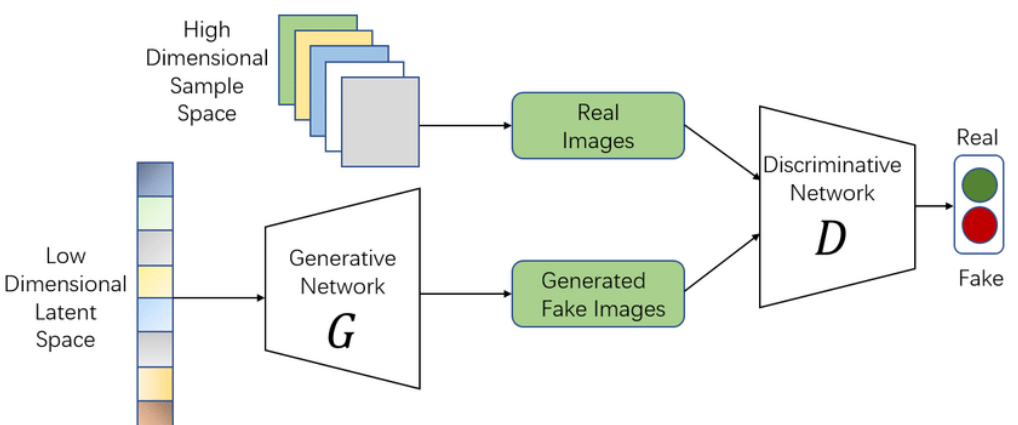


Part II

GENERATIVE ADVERSARIAL NETWORK (GAN)

Generative Adversarial Network

- Generator (or decoder) G takes a latent sampled from a unit Gaussian as input and generates a synthetic image.
- Discriminator D takes an image as input and classifies it as either real or fake (generated).



Objective Function

- A min-max optimization problem, which is known as very difficult to solve

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z\sim p(z)}[\log(1 - D(G(z)))]$$

Sample x from real data distribution

Sample latent code z from Gaussian distribution

$$\min_G \max_D V(D,G) = E_{x\sim p_{data}(x)}[\log D(x)] + E_{z\sim p_z(z)}[\log(1 - D(G(z)))]$$

D should maximize $V(D,G)$

Maximum when $D(x) = 1$

Maximum when $D(G(z)) = 0$

G is independent of this part

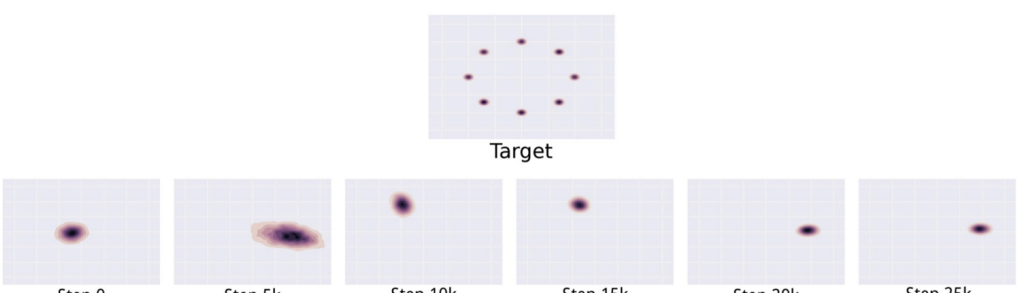
$$\min_G \max_D V(D,G) = \cancel{E_{x\sim p_{data}(x)}[\log D(x)]} + E_{z\sim p_z(z)}[\log(1 - D(G(z)))]$$

G should minimize $V(D,G)$

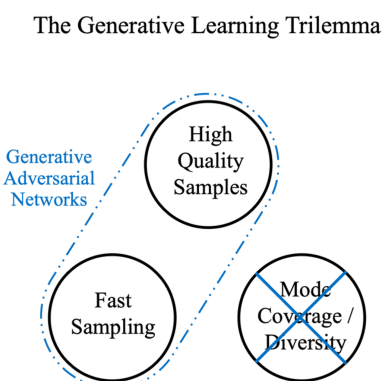
Minimum when $D(G(z)) = 1$

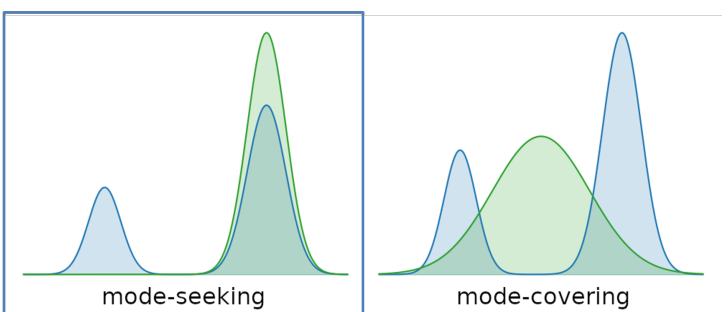
Challenges in GAN Training

1. **Gradient Vanishing**
   - The discriminator becomes too strong and can easily distinguish between real and fake samples.
   - This leads to near-zero gradients, halting the generator's learning process.
2. **Mode Collapse**
   - The generator tends to collapse to a few modes of the data distribution instead of capturing the full diversity of the true data distribution.



SUMMARY OF GAN





Part III

VARIATIONAL AUTOENCODER (VAE)

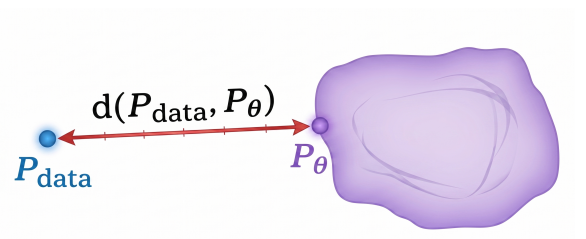
Maximum Likelihood Learning

We want p_θ close to p_{data}. Using KL divergence as the measure:

$$D_{\text{KL}}(p_{\text{data}} \,\|\, p_\theta) = \underbrace{\mathbb{E}_{x\sim p_{\text{data}}}[\log p_{\text{data}}(x)]}_{\text{const w.r.t. } \theta} - \mathbb{E}_{x\sim p_{\text{data}}}[\log p_\theta(x)]$$

$$\min_\theta \; D_{\text{KL}}(p_{\text{data}} \,\|\, p_\theta) \iff \max_\theta \frac{1}{N}\sum_{i=1}^{N} \log p_\theta(x_i) \quad \textbf{(MLE)}$$

$\Longrightarrow$ training = maximizing $\log p_\theta(x)$.

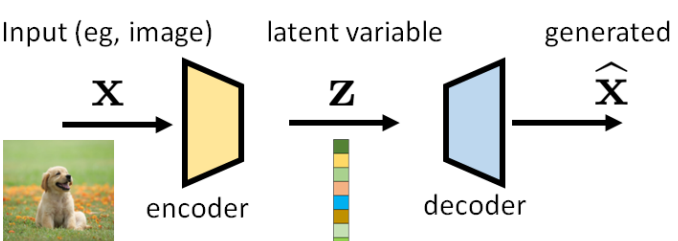


VARIATIONAL AUTOENCODER

The model consists of two components:

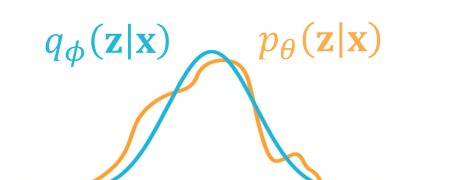
- **Encoder** $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z};\, \mu_\phi(\mathbf{x}),\, \boldsymbol{\sigma}^2_\phi(\mathbf{x})\mathbf{I})$: maps input $\mathbf{x}$ to a distribution over the latent space (mean μ and variance $\boldsymbol{\sigma}^2$).
- **Decoder** $p_\theta(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x};\, D_\theta(\mathbf{z}),\, \sigma^2\mathbf{I})$: maps a sampled latent vector $\mathbf{z}$ back to the data space, reconstructing $\hat{\mathbf{x}}$.

$$p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x}, \mathbf{z})\, d\mathbf{z}$$



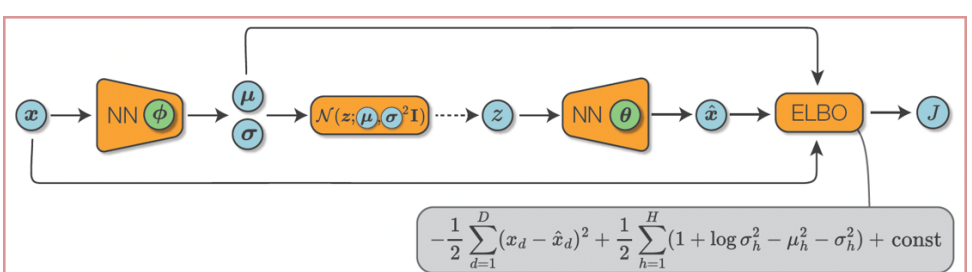
Objective Function: ELBO Derivation

$$
\begin{aligned}
\log p_\theta(\mathbf{x}) = \int_{\mathbf{z}} q_\phi(\mathbf{z}|\mathbf{x}) \log p_\theta(\mathbf{x}) d\mathbf{z} &= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x})] \\
&= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_\theta(\mathbf{x},\mathbf{z})}{p_\theta(\mathbf{z}|\mathbf{x})}\right] \\
&= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_\theta(\mathbf{x},\mathbf{z})}{p_\theta(\mathbf{z}|\mathbf{x})} \cdot \frac{q_\phi(\mathbf{z}|\mathbf{x})}{q_\phi(\mathbf{z}|\mathbf{x})}\right] \\
&= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_\theta(\mathbf{x},\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right] + \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p_\theta(\mathbf{z}|\mathbf{x})}\right] \\
&= \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_\theta(\mathbf{x},\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right]}_{\text{ELBO}} + D_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}|\mathbf{x}))
\end{aligned}
$$



Objective Function: ELBO Decomposition

$$\begin{aligned}
\text{ELBO} = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_\theta(\mathbf{x},\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right] &= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_\theta(\mathbf{x}|\mathbf{z})\, p_\theta(\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right] \\
&= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})] - D_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p_\theta(\mathbf{z})) \\
&\approx \underbrace{-\frac{1}{2}\sum_{d=1}^{D}(x_d - \hat{x}_d)^2}_{\text{reconstruction term}} + \underbrace{\frac{1}{2}\sum_{h=1}^{H}\left(1 + \log \sigma_h^2 - \mu_h^2 - \sigma_h^2\right)}_{\text{prior matching term}} + \text{const}
\end{aligned}$$

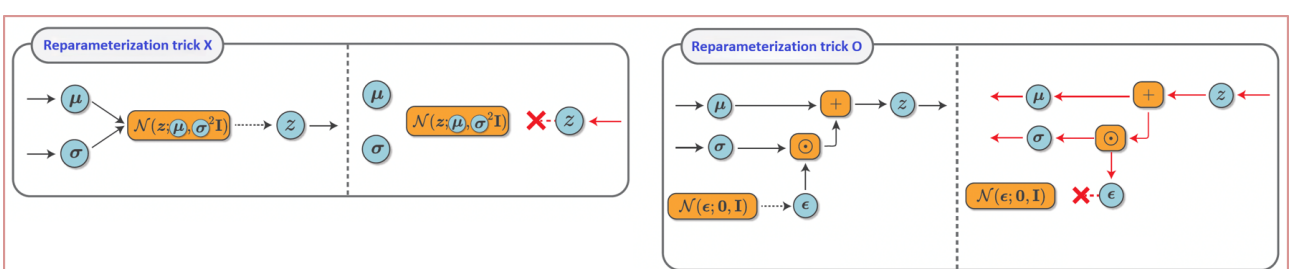


Reparameterization Trick

Stochastic sampling procedure $\rightarrow$ Non-differentiable

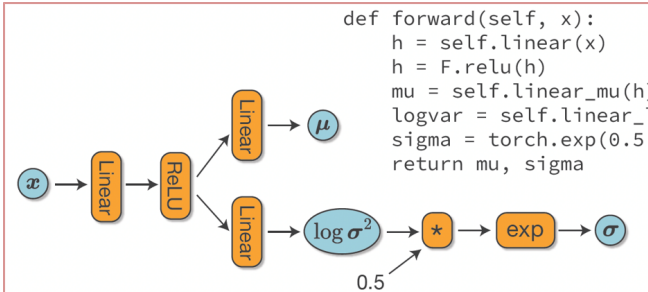
$$\mathbf{z} \sim \mathcal{N}(\mu, \boldsymbol{\sigma}^2\mathbf{I}) \quad \Rightarrow \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \mathbf{z} = \mu + \boldsymbol{\sigma} \odot \epsilon = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_H \end{bmatrix} + \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \vdots \\ \sigma_H \end{bmatrix} \odot \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_H \end{bmatrix}$$

$$\mathbb{E}[z_i] = \mu_i + \sigma_i \, \mathbb{E}[\epsilon_i] = \mu_i, \qquad \mathrm{Var}(z_i) = \sigma_i^2 \, \mathrm{Var}(\epsilon_i) = \sigma_i^2$$



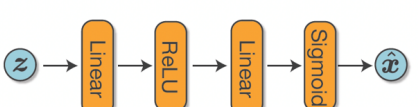
Implementation of VAE

Encoder




```
def forward(self, x):
    h = self.linear(x)
    h = F.relu(h)
    mu = self.linear_mu(h)
    logvar = self.linear_logvar(h)
    sigma = torch.exp(0.5 * logvar)
    return mu, sigma
```

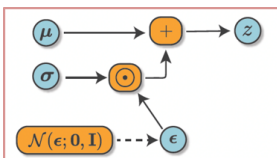
Dcoder




```
def forward(self, z):
    h = self.linear1(z)
    h = F.relu(h)
    h = self.linear2(h)
    x_hat = F.sigmoid(h)
    return x_hat
```

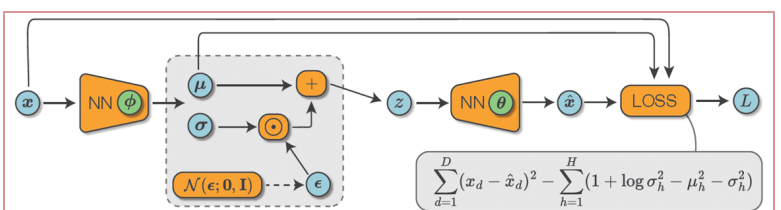
Implementation of VAE

Reparameterization Trick

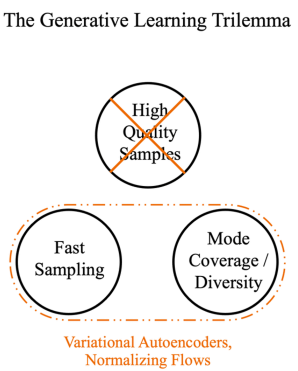


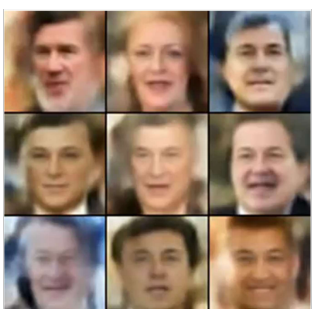

```
def reparameterize(mu, sigma):
    eps = torch.randn_like(sigma)
    z = mu + eps * sigma
    return z
```

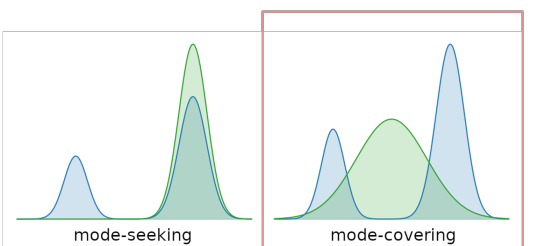
Loss Function



SUMMARY OF VAE







Part IV

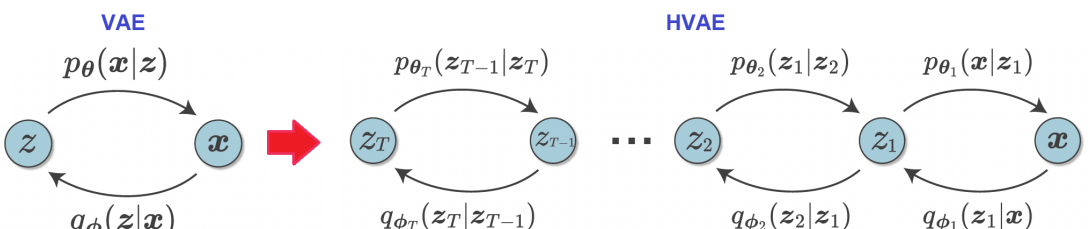
Hierarchical Variational Autoencoder (HVAE)

Extending VAE to Multiple Latent Layers

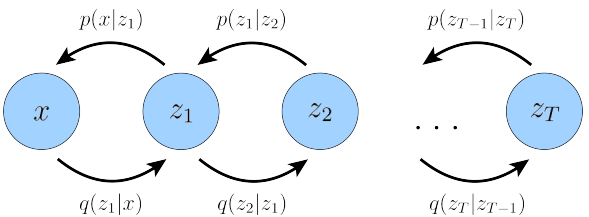
VAE uses a single latent layer $\mathbf{z}$. Two structural limitations:

- A single $\mathbf{z}$ must simultaneously encode all levels of abstraction (shape, texture, identity, ...).
- The decoder reconstructs $\mathbf{x}$ from $\mathbf{z}$ in a single step.

A natural extension: stack multiple latent layers $\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_T$, each capturing structure at a different level of abstraction.



HIERARCHICAL VARIATIONAL AUTOENCODER



Markov assumption: each layer depends only on its adjacent layer.
Forward process (encoder):

$$q_\phi(\mathbf{x}_{1:T}|\mathbf{x}) = q_{\phi_1}(\mathbf{x}_1|\mathbf{x}) \prod_{t=2}^{T} q_{\phi_t}(\mathbf{x}_t|\mathbf{x}_{t-1})$$

Reverse process (decoder):

$$p_\theta(\mathbf{x}, \mathbf{x}_{1:T}) = p(\mathbf{x}_T) \prod_{t=1}^{T} p_{\theta_t}(\mathbf{x}_{t-1}|\mathbf{x}_t)$$

- $p(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I})$: simple prior
- Each encoder q_{ϕ_t} and decoder p_{θ_t} is a learnable neural network

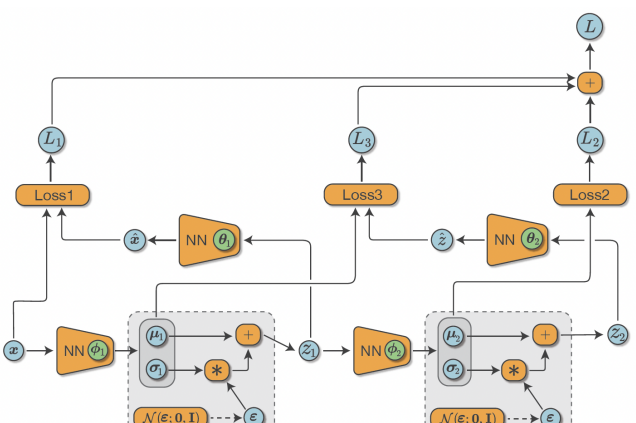
Objective Function: ELBO

The same variational strategy as VAE applies.

$$
\begin{aligned}
\log p_\theta(\mathbf{x}_0) &= \log \int p_\theta(\mathbf{x}_{0:T})\, d\mathbf{x}_{1:T} \\
&= \log \int \frac{p_\theta(\mathbf{x}_{0:T})}{q_\phi(\mathbf{x}_{1:T}|\mathbf{x}_0)} \cdot q_\phi(\mathbf{x}_{1:T}|\mathbf{x}_0)\, d\mathbf{x}_{1:T} \\
&= \log \mathbb{E}_{q_\phi}\left[\frac{p_\theta(\mathbf{x}_{0:T})}{q_\phi(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] \\
&\geq \underbrace{\mathbb{E}_{q_\phi}\left[\log \frac{p_\theta(\mathbf{x}_{0:T})}{q_\phi(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right]}_{\text{ELBO}} \\
&= \underbrace{\mathbb{E}_{q_\phi}[\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]}_{\mathcal{L}_0\text{: reconstruction}} - \underbrace{D_{\mathrm{KL}}(q_\phi(\mathbf{x}_T|\mathbf{x}_{T-1}) \,\|\, p(\mathbf{x}_T))}_{\mathcal{L}_T\text{: prior matching}} - \sum_{t=1}^{T-1} \underbrace{\mathbb{E}_{q_\phi} D_{\mathrm{KL}}\left(q_\phi(\mathbf{x}_t|\mathbf{x}_{t-1}) \,\middle\|\, p_\theta(\mathbf{x}_t|\mathbf{x}_{t+1})\right)}_{\mathcal{L}_{t-1}\text{: consistency}}
\end{aligned}
$$

Two-Level HVAE

$$\begin{aligned}\text{ELBO} &= \mathbb{E}_{q_{\phi_1}}\left[\log p_{\theta_1}(\mathbf{x}|\mathbf{x}_1)\right] - \mathbb{E}_{q_{\phi_1}}\left[D_{\text{KL}}\left(q_{\phi_2}(\mathbf{x}_2|\mathbf{x}_1) \,\|\, p(\mathbf{x}_2)\right)\right] - \mathbb{E}_{q_{\phi_2}}\left[D_{\text{KL}}\left(q_{\phi_1}(\mathbf{x}_1|\mathbf{x}) \,\|\, p_{\theta_2}(\mathbf{x}_1|\mathbf{x}_2)\right)\right] \\ &\approx -\frac{1}{2}\sum_{d=1}^{D}(\mathbf{x}_d - \hat{x}_d)^2 + \frac{1}{2}\sum_{h=1}^{H}\left(1 + \log\sigma_{2,h}^2 - \mu_{2,h}^2 - \sigma_{2,h}^2\right) + \frac{1}{2}\sum_{h=1}^{H}\left(1 + \log\sigma_{1,h}^2 - (\mu_{1,h} - \hat{z}_{1,h})^2 - \sigma_{1,h}^2\right)\end{aligned}$$



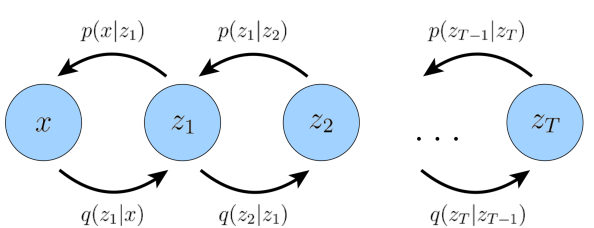
Part V

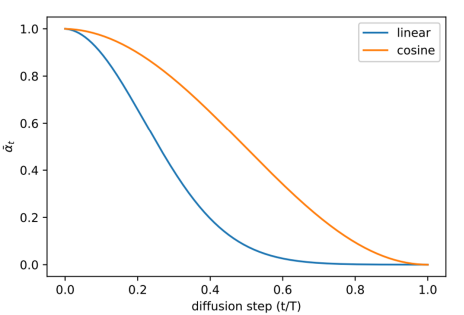
Denoising Diffusion Probabilistic Models (DDPM)

HVAE $\rightarrow$ DDPM

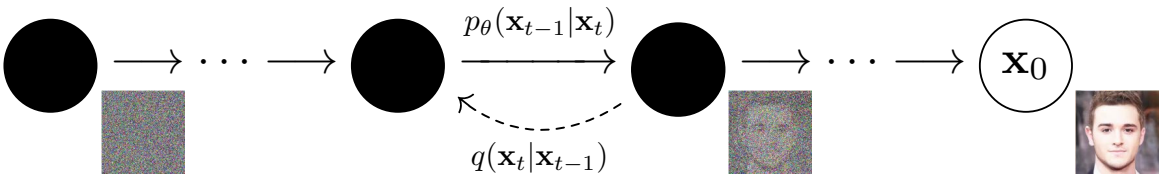
DDPM is a Markovian HVAE with three key restrictions:

1. **Data dimension = Latent dimension** $\Rightarrow \dim(\mathbf{x}) = \dim(\mathbf{x}_t)$ for all t
2. **The encoder is fixed — not learned**
   - A pre-defined linear Gaussian transition
   $$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t;\ \sqrt{\alpha_t}\,\mathbf{x}_{t-1},\ (1-\alpha_t)\mathbf{I})$$
3. **The final latent follows a standard Gaussian**
   $$\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$





Forward & Reverse Process



- **Forward Process:** The process of gradually adding Gaussian noise to $\mathbf{x}_0$ to produce $\mathbf{x}_T$, which is fully noised.
- **Reverse Process:** The process of generating an image by gradually removing noise from $\mathbf{x}_T$.
  - When each forward step is sufficiently small, the reverse step $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is also approximately Gaussian.

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1};\ \mu_\theta(\mathbf{x}_t, t),\ \sigma_t^2\mathbf{I})$$

Forward Process

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t;\ \sqrt{\alpha_t}\,\mathbf{x}_{t-1},\ (1-\alpha_t)\mathbf{I})$$

$$\begin{aligned}
\mathbf{x}_t &= \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\epsilon_{t-1} \\
&= \sqrt{\alpha_t}(\sqrt{\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_{t-1}}\epsilon_{t-2}) + \sqrt{1-\alpha_t}\epsilon_{t-1} \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{\alpha_t - \alpha_t\alpha_{t-1}}\epsilon_{t-2} + \sqrt{1-\alpha_t}\epsilon_{t-1} \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\epsilon_{t-2} \\
&\ \ \vdots \\
&= \sqrt{\prod_{i=1}^{t}\alpha_i}\mathbf{x}_0 + \sqrt{1-\prod_{i=1}^{t}\alpha_i}\epsilon_0
\end{aligned}$$

Define $\bar{\alpha}_t := \prod_{i=1}^{t}\alpha_i$.

$$\boxed{\begin{aligned}
q(\mathbf{x}_t|\mathbf{x}_0) &= \mathcal{N}(\mathbf{x}_t;\ \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0,\ (1-\bar{\alpha}_t)\mathbf{I}) \\
\mathbf{x}_t &= \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon, \qquad \epsilon \sim \mathcal{N}(\mathbf{0},\mathbf{I})
\end{aligned}}$$

DDPM ELBO

Applying the Markovian HVAE ELBO to DDPM:

$$\text{ELBO} = \underbrace{\mathbb{E}_q[\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]}_{\mathcal{L}_0\text{: reconstruction}} - \underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \,\|\, p(\mathbf{x}_T))}_{\mathcal{L}_T\text{: } \approx 0} - \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}\left(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \,\middle\|\, p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)\right)}_{\mathcal{L}_{t-1}\text{: denoising matching}}$$

- Since the forward process is fixed (not learned), the encoder is no longer q_ϕ but simply q.
- $\mathcal{L}_T \approx 0$: noise schedule designed so that $\bar{\alpha}_T \approx 0$.

Forward Posterior $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$

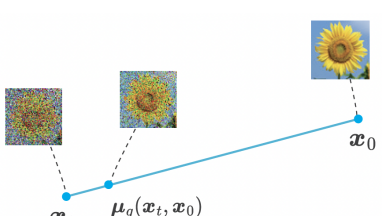
Since $q(\mathbf{x}_t|\mathbf{x}_{t-1})$, $q(\mathbf{x}_{t-1}|\mathbf{x}_0)$, and $q(\mathbf{x}_t|\mathbf{x}_0)$ are all Gaussian, Bayes' rule gives a closed-form Gaussian:

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})\, q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} = \mathcal{N}(\mathbf{x}_{t-1};\ \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0),\ \tilde{\beta}_t \mathbf{I})$$

$$\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t}\mathbf{x}_0, \qquad \tilde{\beta}_t = \frac{(1-\bar{\alpha}_{t-1})(1-\alpha_t)}{1-\bar{\alpha}_t}$$

The denoising matching term simplifies to mean matching (both distributions are Gaussian with fixed variance):

$$\mathcal{L}_{t-1} \propto \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left[\|\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|^2\right]$$



THREE PREDICTION TARGETS

$$\tilde{\mu}_t = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t}\mathbf{x}_0, \qquad \mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon$$

- Mean predictor: $\mathcal{L} = \mathbb{E}\left[\|\tilde{\mu}_t - \mu_\theta(\mathbf{x}_t, t)\|^2\right]$
- Original-sample predictor: $\mathcal{L} = \mathbb{E}\left[\|\mathbf{x}_0 - \hat{\mathbf{x}}_\theta(\mathbf{x}_t, t)\|^2\right]$
- Noise predictor: $\mathcal{L} = \mathbb{E}\left[\|\epsilon - \hat{\epsilon}_\theta(\mathbf{x}_t, t)\|^2\right]$

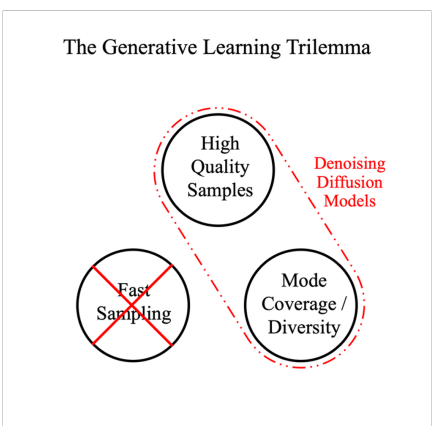
SUMMARY OF DDPM

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1, \dots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on
$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$
6: **until** converged

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \dots, 1$ **do**
3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$



Improving Sampling Speed

The primary drawback of DDPM is its slow sampling speed (requiring $T \approx 1000$ steps).
Various approaches have been proposed to achieve fast sampling:

1. **Reducing Sampling Steps (Advanced Solvers)**
   - Uses advanced numerical solvers (e.g., ODE solvers) to enable larger step sizes.
2. **Knowledge Distillation**
   - Trains a fast "student" model to mimic the multi-step generation trajectory of a slow "teacher" diffusion model.
3. **Latent Diffusion Models**
   - Compresses high-dimensional images into a smaller latent space and performs the diffusion process there.

CONTROLLABILITY: GUIDANCE

Guidance is a technique to steer the generative process toward a specific condition y (e.g., a text prompt or class label).

- **Classifier Guidance**
  - Uses the gradient of a separate pre-trained classifier $p(y|\mathbf{x}_t)$ to guide the reverse step.
  - $\hat{\epsilon}_\theta(\mathbf{x}_t, y) \approx \epsilon_\theta(\mathbf{x}_t) - w\sqrt{1-\bar{\alpha}_t}\nabla_{\mathbf{x}_t} \log p(y|\mathbf{x}_t)$
- **Classifier-Free Guidance (CFG)**
  - Most widely used. It jointly trains a conditional and an unconditional model (using null-conditioning $\emptyset$).
  - Predicts the noise as a linear combination:

$$\hat{\epsilon}_\theta(\mathbf{x}_t, y) = \epsilon_\theta(\mathbf{x}_t, \emptyset) + w \cdot (\epsilon_\theta(\mathbf{x}_t, y) - \epsilon_\theta(\mathbf{x}_t, \emptyset))$$

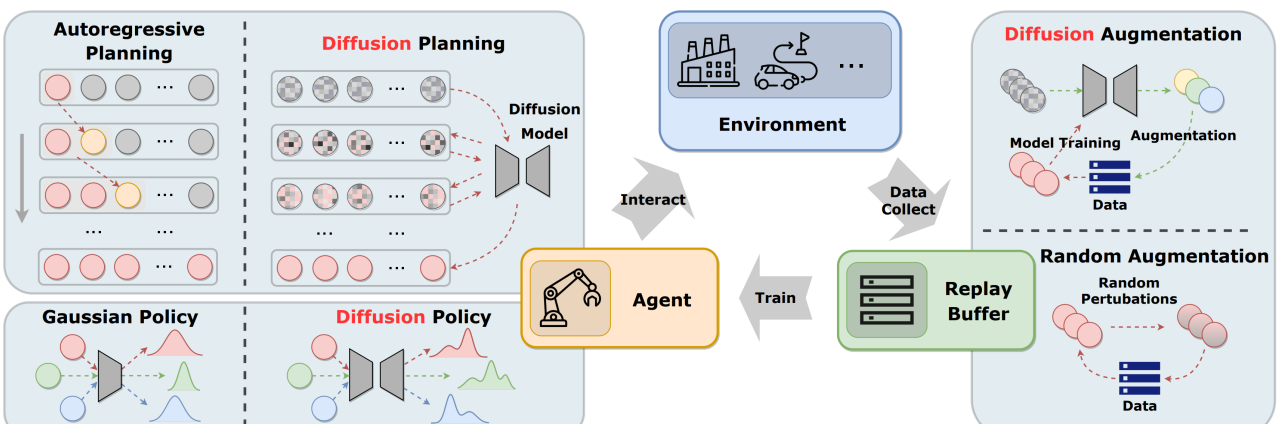
- **Fidelity vs. Diversity Trade-off:**
  - A higher guidance scale w improves **fidelity** (alignment with the prompt) but reduces **sample diversity**.

Roles of Diffusion Models in Reinforcement Learning

Role	Generated Object
Planner	Trajectory $\tau = (s_1, a_1, \ldots, s_H, a_H)$
Policy	Action a
Data Synthesizer	Transition (s, a, r, s')

- **Planner:** Generates long-horizon trajectories as a single sample, capturing complex global dependencies.
- **Policy:** Expresses multi-modal action distributions, overcoming the limitations of Gaussian policies
- **Data Synthesizer:** Acts as a world model or for data augmentation to improve sample efficiency

Roles of Diffusion Models in Reinforcement Learning



Thank you